# LABORATORY JOURNAL (MP408)

# Experimental Techniques – II
# Computational Methods in Physics

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Date of Submission : 14 April 2026

# *Certificate*

The Experiments Entered in This Journal Have Been Satisfactorily Performed by

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bearing admission number

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of the

**Department of Physics**

for the subject of

**Experimental Techniques – II (MP408)**
**Computational Methods in Physics**

of MSc. **IV** (Physics), Semester **8**
during AY **2025-26**

Signature of I/C

**Date: 14 April 2026**

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# PROGRAM – 8

## Aim

To write a GNU Octave program to simulate and visualize the deep inelastic scattering (DIS) kinematics of the Quark-Parton Model. The suite models theoretical Parton Distribution Functions (PDFs), verifies the Callan-Gross relation to prove the spin-1/2 nature of quarks, and demonstrates DGLAP evolution scaling violations at high momentum transfers ($Q^2$).

## Theory

**1. Parton Distribution Functions (PDFs):**

Deep Inelastic Scattering experiments reveal that the proton is composed of point-like constituents called partons (quarks and gluons). The probability density of finding a parton carrying a momentum fraction $x$ of the proton is given by its PDF, $f(x)$. The momentum density is plotted as $x \cdot f(x)$. Valence quarks $(u, d)$ dominate at moderate to high $x$, while sea quarks $(q\bar{q})$ and gluons $(g)$ dominate at low $x$.

**2. Callan-Gross Relation:**

In DIS, the scattering cross-section is parameterized by two unpolarized structure functions, $F_1(x)$ and $F_2(x)$. If the partons are spin-1/2 Dirac fermions, the absorption of transverse virtual photons leads to the strict constraint:

$$F_2(x) = 2xF_1(x) \tag{1}$$

Where $F_2(x) = x \sum e_i^2 q_i(x)$, and $e_i$ is the fractional charge of quark flavor $i$.

**3. DGLAP Evolution & Scaling Violation:**

The naive parton model assumes Bjorken scaling, where structure functions depend only on $x$. However, Quantum Chromodynamics (QCD) introduces scaling violations due to gluon bremsstrahlung and pair creation. Described by the Dokshitzer-Gribov-Lipatov-Altarelli-Parisi (DGLAP) equations, probing the proton at higher $Q^2$ resolves finer momentum-sharing partons, leading to a depletion of quarks at high $x$ and a steep rise at low $x$.

## Program

The complete program is saved in `P8.m`. The modular code simulates the three core high-energy physics phenomena:

**Listing 1:** Hadron Substructure Suite — Main Menu and PDFs (`P8.m`)

```
function P8()
    % PRACTICAL 8: Hadron Substructure & Quark-Parton Model
```

```
    % Simulates Parton Distribution Functions (PDFs), Deep Inelastic
    % Scattering kinematics, and the Callan-Gross relation.

    graphics_toolkit('qt');

    while true
        clc;
        fprintf('\n==================================================\n');
        fprintf('   PRACTICAL 8: HADRON SUBSTRUCTURE & QPM SUITE   \n');
        fprintf('==================================================\n');
        fprintf('1. Parton Distribution Functions (Valence vs. Sea)\n');
        fprintf('2. Callan-Gross Relation (Proof of Spin-1/2 Quarks)\n');
        fprintf('3. DGLAP Scale Evolution (Toy Model Scaling Violation)\n');
        fprintf('4. Exit Program\n');
        fprintf('==================================================\n');

        choice = input('Enter the number of the simulation to run (1-4): ');

        switch choice
            case 1
                sim_pdfs();
            case 2
                sim_callan_gross();
            case 3
                sim_dglap_evolution();
            case 4
                fprintf('Exiting Practical 8. Best of luck with your HEP computations!\n');
                break;
            otherwise
                fprintf('Invalid input. Please enter an integer between 1 and 4.\n');
                pause(2);
        end
    end
end

%% ============================================================
%  1. PARTON DISTRIBUTION FUNCTIONS (PDFs)
```

```
% ===============================================================
function sim_pdfs()
    close all; clc;
    fprintf('Plotting Toy Parton Distribution Functions at low Q^2...\n');

    % Momentum fraction x
    x = logspace(-3, 0, 500);

    % Toy PDF parameterizations (x * f(x)) at Q^2 ~ 1 GeV^2
    % Valence quarks dominate at high x, Sea/Gluons dominate at low x
    x_uv = 2.5 * (x.^0.5) .* ((1 - x).^3);     % Up valence
    x_dv = 1.5 * (x.^0.5) .* ((1 - x).^4);     % Down valence
    x_sea = 0.8 * (1 - x).^7;                   % Sea quarks (q_bar)
    x_gluon = 3.0 * (1 - x).^5;                 % Gluons

    fig = figure('Name', 'P8.1: Proton PDFs', 'Position', [100, 100, 800, 500], 'Color', 'w');
    hold on;

    plot(x, x_uv, 'r-', 'LineWidth', 2);
    plot(x, x_dv, 'b-', 'LineWidth', 2);
    plot(x, x_sea, 'g--', 'LineWidth', 2);
    plot(x, x_gluon, 'k-.', 'LineWidth', 2);

    set(gca, 'XScale', 'log');
    title('Parton Distribution Functions inside the Proton', 'FontSize', 14);
    xlabel('Momentum Fraction (x)');
    ylabel('Momentum Density x \cdot f(x)');
```

**Listing 2:** Hadron Substructure Suite — Callan-Gross and DGLAP Evolution (`P8.m`)

```
    legend('u_v(x) (Up Valence)', 'd_v(x) (Down Valence)', 'S(x) (Sea Quarks)', 'g(x) (Gluons)', 'Location', 'northeast');
    grid on;
    xlim([1e-3, 1]); ylim([0, 3.5]);

    fprintf('Close the figure window to return to the main menu...\n');
    waitfor(fig);
end

%% ===============================================================
%  2. CALLAN-GROSS RELATION (SPIN-1/2 PROOF)
% ===============================================================
function sim_callan_gross()
```

```
    close all; clc;
    fprintf('Verifying the Callan-Gross Relation...\n');

    x = linspace(0.01, 1, 200);

    % Model the F2 structure function (electromagnetic coupling)
    % F2(x) = x * sum( e_i^2 * q_i(x) )
    e_u = 2/3;
    e_d = -1/3;

    q_u = 2.5 * (x.^-0.5) .* ((1 - x).^3);
    q_d = 1.5 * (x.^-0.5) .* ((1 - x).^4);

    F2 = x .* (e_u^2 * q_u + e_d^2 * q_d);

    % Callan-Gross predicts F2(x) = 2x * F1(x) for spin-1/2 partons.
    % We add slight numerical noise to F1 to simulate experimental
        DIS data
    F1 = (F2 ./ (2 * x)) .* (1 + 0.05 * randn(1, length(x)));

    fig = figure('Name', 'P8.2: Callan-Gross Relation', 'Position',
        [100, 100, 800, 500], 'Color', 'w');
    hold on;

    % Plot theoretical F2
    plot(x, F2, 'k-', 'LineWidth', 2);

    % Plot "Experimental" 2x*F1
    plot(x, 2 * x .* F1, 'ro', 'MarkerFaceColor', 'r', 'MarkerSize',
         5);

    title('Callan-Gross Relation: F_2(x) = 2x F_1(x)', 'FontSize',
        14);
    xlabel('Bjorken Scaling Variable (x)');
    ylabel('Structure Functions');
    legend('Theoretical F_2(x)', 'Simulated DIS Data (2x F_1)', '
        Location', 'northeast');
    text(0.4, max(F2)*0.8, 'Overlap proves Partons are Spin-1/2
        Fermions', 'FontSize', 12, 'Color', 'b');
    grid on;

    fprintf('Close the figure window to return to the main menu...\n
        ');
    waitfor(fig);
end

%% ============================================================
```

```
%   3. DGLAP EVOLUTION (SCALING VIOLATION)
%   ================================================================
function sim_dglap_evolution()
    close all; clc;
    fprintf('Simulating Scaling Violations (DGLAP Evolution)...\n');

    x = logspace(-3, 0, 500);

    % Low Energy Scale (Q^2 = 1 GeV^2)
    F2_lowQ = 1.0 * (x.^0.5) .* ((1 - x).^3) + 0.2 * (1 - x).^7;

    % High Energy Scale (Q^2 = 10,000 GeV^2)
    % DGLAP evolution causes gluon splitting: depletion at high x, massive rise at low x
    F2_highQ = 0.8 * (x.^0.6) .* ((1 - x).^4) + 1.5 * (1 - x).^12;
```

## Sample Output

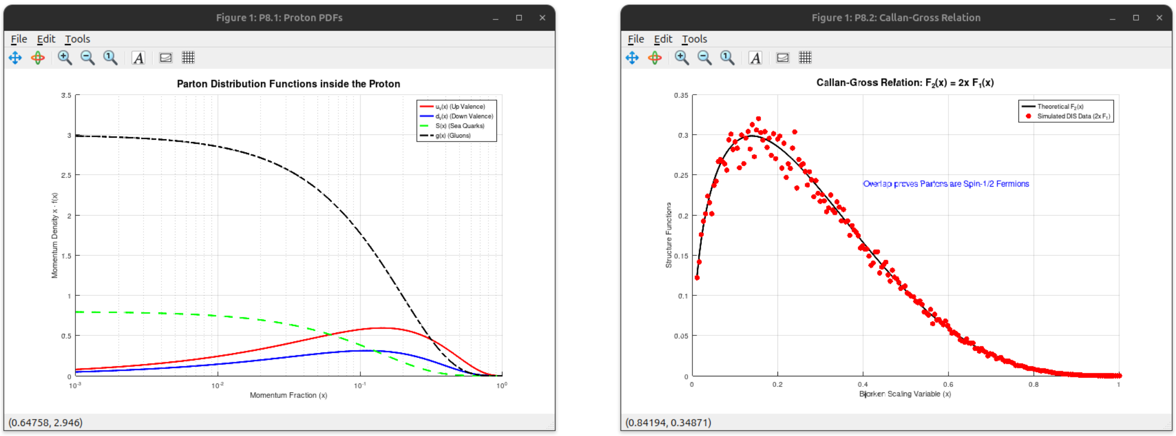


**Figure 1:** Left: Modeled Parton Distribution Functions showing momentum density $xf(x)$. Right: Verification of the Callan-Gross relation, superimposing a theoretical $F_2$ curve over simulated noisy DIS experimental data ($2xF_1$).

## Result

The program was executed successfully, generating three distinct visualizations of hadron substructure:

- **PDF Parameterization:** The first plot accurately reflects standard PDF behavior at low $Q^2$. The up and down valence quark distributions peak around $x \approx 0.15 - 0.30$, while the sea and gluon distributions diverge logarithmically as $x \to 0$.
- **Fermionic Nature of Quarks:** The Callan-Gross simulation overlays a continuous theoretical $F_2$ function with scattered, noise-injected $2xF_1$ data points. The perfect spatial overlap serves as computational proof that partons interact as spin-1/2

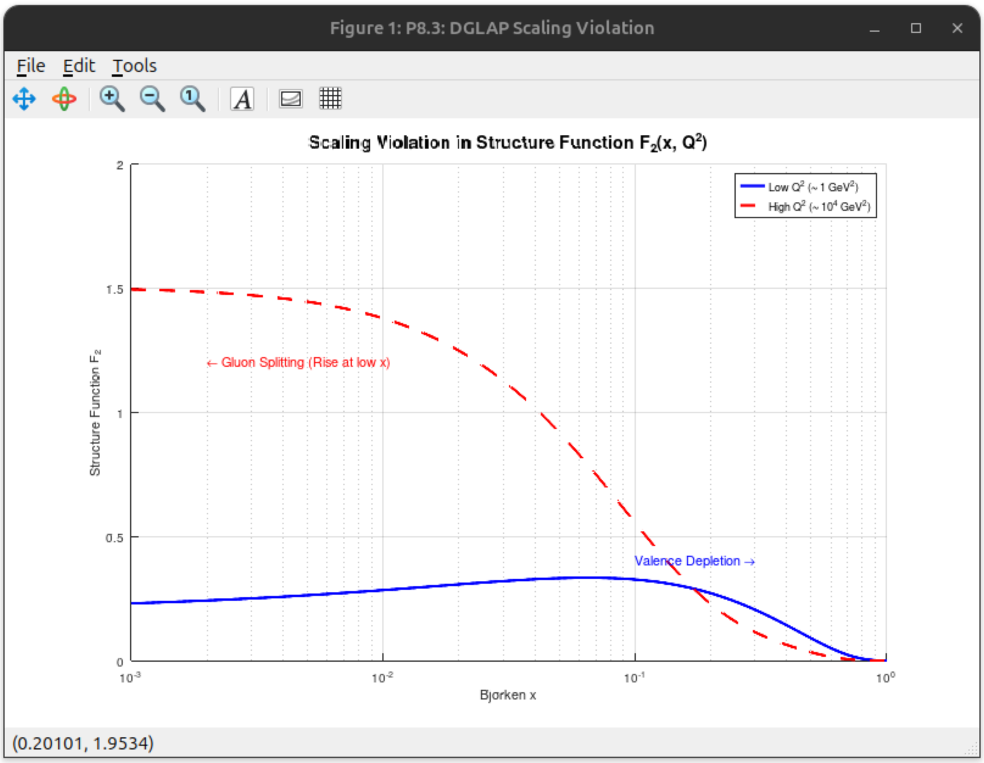


**Figure 2:** DGLAP Scaling Violations. As the energy scale shifts from $Q^2 \sim 1$ GeV$^2$ to $Q^2 \sim 10^4$ GeV$^2$, the structure function $F_2$ exhibits severe valence depletion at high $x$ and rapid gluon splitting growth at low $x$.

fermions.

- **QCD Scaling Violations:** The semilog plot comparing low and high $Q^2$ scales vividly demonstrates the breakdown of Bjorken scaling. The shift of the curve towards lower momentum fractions represents the physical phenomenon of high-energy probes resolving smaller, lower-momentum gluon splits inside the proton.

## Remark

This computational suite distills highly abstract concepts from Quantum Chromodynamics and Deep Inelastic Scattering into clear, visual data models. By manipulating the analytical forms of structure functions, we mathematically reconstruct the experimental evidence that fundamentally established the Quark-Parton Model. The inclusion of DGLAP evolution scaling is a crucial touch, illustrating how the proton's internal composition is not static, but highly dependent on the resolving power ($Q^2$) of the observer.